INDEPENDENT, ALTERNATIVE, AND COMMUNITY MEDIA AN OPEN READER

Edited and with an introduction by
David W. Park & Meghan Grosse

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Independent, Alternative, and Community Media: An Open Reader

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CHAPTER 29

‘Let’s Keep Music Special. F—Spotify’: On-Demand Streaming and the Controversy over Artist Royalties

Lee Marshall

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Abstract

On-demand streaming music services have expanded significantly in recent years. Services such as Spotify and Deezer are widely expected to become the dominant means of mass music consumption in the future and have been significant factors in helping recording industries in some countries arrest their long-term decline. The emergence of such services (especially the largest of them, Spotify) has caused controversy, however, with both major stars and smaller independent artists publicly criticising the low levels of royalty payments such services have thus far generated. This paper provides an overview of the controversy, concentrating on the criticisms of Spotify from independent labels and musicians. In such a rapidly moving sector, it is unwise to offer any definitive predictions and thus the purpose of this paper is to outline as clearly as possible the parameters of the current debate. To achieve this, it considers the business models associated with streaming music and their place within the contemporary recorded music ecosystem, most notably in relation to piracy and declin-

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ing record purchases. It concludes with a consideration of streaming services in relation to the broader power dynamics of the recording industry suggesting that while services like Spotify may cause problems for independent musicians they rely upon a logic of scaling that correlates with the existing practices of the major record labels.

Introduction

One thing that the history of popular music tells us is that the introduction of significant new technologies is rarely uncontroversial. New inventions, such as the microphone (Frith 1986) or the sampler (Theberge 2004) get challenged and ridiculed by parties whose interests may be threatened by new social practices of music-making and music-consuming. A recent example of this kind of conflict is the controversy over the amount of payment being given to artists for allowing their music to be made available on streaming services such as Deezer and, especially, Spotify. It is a controversy that periodically appears in the mainstream media, such as Radiohead singer Thom Yorke's proclamation that Spotify is the 'last gasp of the old industry… the last desperate fart of a dying corpse' (Dredge 2013) or Taylor Swift's decision to remove all of her back catalogue from the service upon the release of her 2014 album, 1989 (Dickey 2014).

The controversy is not merely a few superstars grumbling, however, as the majority of concerns have actually been raised by independent artists several rungs down the music industry ladder. While such discontent could perhaps be assumed to be merely an economic spat about how the pie should be divided, it actually reflects more deep-seated uncertainties regarding the changing nature of the music commodity and its impact upon the perceived value of music, as well as more long-standing issues regarding the dominant power relations within the music industry. This latter point is notable because it contradicts a common perception that the disintermediating nature of the internet should be a boon to musicians beyond the major label system (for example, Baym 2011; Wikstrom 2009), with the emergence of 'long tail' economics (Anderson 2007) enabling unsigned artists to sell their music to a wide audience. However, the seemingly inexorable shift to streaming services has the potential to be a way through which old industry structures reassert themselves. Yorke is thus correct to associate Spotify with the 'old industry' but perhaps premature when describing it as a 'dying corpse'.

An introduction to streaming services

Before outlining details of the controversy, it is important to differentiate streaming music services. Often generically referred to as 'music in the cloud' (Morris 2011), streaming services can be split into three types. The first is streaming radio (sometimes referred to as webcasting). This can simply be an online version of existing radio services, or online-only radio stations that specialise in particular genres or localities, but the more distinctive form is personalised, in that the service selects songs to play for each individual listener based on previous listening preferences. This kind of streaming radio service thus provides a more bespoke experience than broadcast radio but, in comparison to other kinds of streaming service, the important characteristic is that it is non-interactive: the listener does not choose what songs will play. The most established streaming radio service is Pandora, which announced in April 2013

that it had 200 million registered users in the United States, 70 million of which were 'active' each month (Peoples 2013).[1] The majority of Pandora's income comes from advertising but it also has 2.5 million subscribers paying a monthly fee to avoid ads and listening limits. Another prominent player in this field is iHeartRadio, a mixture of traditional radio services and Pandora-like tailored recommendations, created by terrestrial radio provider Clear Channel,[2] and boasting 50 million users (Solsman 2014).

The second kind of streaming music service are locker services, such as iTunes match and Amazon's Cloudplayer. In these services, users can listen to 'their' music directly from the internet, either songs that they upload directly to the service's servers or that are automatically 'matched' (to, for example, a user's Amazon CD purchases). Locker services thus offer mobile access to a listener's entire MP3 collection. However, they are a relatively small part of the cloud-based music landscape and, in all likelihood, a temporary phenomenon, designed to ease consumer experience until the 'purer' forms of cloud-based services are adopted more widely.

The third kind of streaming music services are 'on-demand'. Such services are similar to streaming radio services, such as Pandora, with one crucial difference: the listener chooses what music is played. They are thus interactive rather than non-interactive. As such, they provide a listening experience similar to locker services, except that choice is not limited to an individual's collection: instead the listener can choose from a vast catalogue of music (upwards of 30 million songs on some services). Spotify has established itself as the most famous of these services but there are other significant providers. The most important of them is Deezer, the second largest on-demand service with 16 million active users and 6 million subscribers (Peoples 2014). It has garnered fewer headlines than Spotify because it has strategically avoided entering the mainstream US market but is the most global service, available in 180 countries. Other services include Rhapsody (until the emergence of Spotify, the largest subscription service in the United States, with 2 million subscribers), US-based Rdio and WiMP, based in Norway.[3]

On-demand services often include radio functions so there is some overlap between services but the most important distinction is the one between non-interactive and interactive services. Within the music industry, it is widely accepted that on-demand services are heralding a significant reconfiguration of recorded music consumption, from being based on ownership of music to being based on access, renting rather than buying records. This acceptance reflects the manner in which streaming music has begun to enter the mainstream in recent years. This can be seen in the global expansion and accelerating growth in user/subscriber figures among the aforementioned services but, probably more significantly, in the way in which the major internet companies are now becoming involved in the on-demand streaming market. For example, Amazon offers a limited streaming service as a benefit to subscribers of its Amazon Prime service, while Google actually provides two streaming subscription services: Google Play Music All Access and YouTube's Music Key service.[4] On top

[1] Pandora was available only in the United States until December 2012, when it launched in Australia and New Zealand and where it now has 2 million listeners.

[2] Clear Channel has subsequently changed its name to iHeartMedia.

[3] WiMP has subsequently been purchased and rebranded as Tidal.

[4] Although often not recognised as such, YouTube is actually the largest music streaming service, and is the de

of this, Apple, a long-time rejecter of streaming music, purchased Beats Music (primarily a headphone manufacturer but with a small on-demand streaming music service) in May 2014 and launched its own on-demand streaming service in June 2015.

The backlash against Spotify

The broader issues discussed in this paper, concerning the changing nature of the musical commodity and the economic reward for recorded music, are applicable to all of the on-demand streaming services. This paper will focus on Spotify, however, given that it is the most prominent of such services and the one which has been the centre of public debate about the levels of remuneration received by artists for streaming. The service was launched from its native Sweden in 2008 and it currently offers a catalogue of 'over 30 million tracks' in 58 countries (Spotify 2014). In January 2015, Spotify announced that the service had achieved 60 million active users and 15 million paying subscribers (Spotify 2015). Its growth in subscriber numbers has been impressive: according to public announcements, it took almost 2 years to get its first 500,000 subscribers but then added 2 million in approximately a year, a further 3.5 million in the following 18 months, 4 million more in the next 14 months and 5 million more in just 7 months. It is growing exponentially and is the poster boy of the new, optimistic, digital record industry—the one that has stopped bleating about piracy and is doing something new and funky. Yet despite, or perhaps because of, its success, Spotify has also received a large amount of criticism, most notably from artists about the levels of remuneration they receive from the service.

The public discussion of Spotify payments first began in 2009, when it was widely reported that Lady Gaga had received a payment of just $167 for her song 'Poker Face', which had been streamed over a million times (for example, Smith 2009).[5] The statistic was misleading[6] but the mud stuck—it is easy to find recent stories repeating the bogus fact (for example, Marsden 2012)—and the idea that Spotify did not adequately reward artists gained public credence. It was around the time that Spotify launched in the United States in 2011 that the controversy surrounding payments became more consistent as a number of smaller US record labels withdrew their catalogues from the service. Seemingly, the first to do so (though not the first to announce it) was Projekt Records, a Brooklyn-based gothic rock label, formed in 1983, which withdrew its catalogue from all on-demand streaming services in July. Label founder Sam Rosenthal stated that these services 'are not a viable way forward for the music industry' and was vocal in his criticism of Spotify:

> In the world I want to live in, I envision artists fairly compensated for their creations… Spotify is NOT a service that does this. Projekt will not be part of this unprincipled concept. For a stream on Spotify… on average $0.0013 is paid to Projekt's Digital Distributor. 5000 plays generates around $6.50. In comparison, 5000 track downloads at iTunes generates $3487. … To earn the U.S. monthly

facto source of music listening for significant numbers of young internet users. It has approximately 140 million weekly music video users (Mulligan 2014a).

[5] For the original Swedish news story, see Andersson (2009).

[6] The amount related to songwriter's publishing royalty only (not the income from the recording itself, which conventionally is about nine times higher), after collecting society deductions, for a song that was co-written, covered only a short period of time just after Spotify had launched and related to streams in Sweden only (see Barnett 2010).

> minimum wage—$1160—892,307 plays a month are needed at Spotify. This is not a viable number for artists. (Rosenthal 2011)

The next label to pull its catalogue out of Spotify, and the first to be publicly reported, was Century Media, a metal label based in Dortmund and California. In August, they announced the withdrawal of its repertoire, along with its associated labels, 'to protect the interests of its artists'. While praising Spotify as 'a great tool to discover new music', Century countered that:

> Physical sales are dropping drastically in all countries where Spotify is active.… Since the artists need to sell their music to continue their creativity, Spotify is a problem for them. This is about survival, nothing less and it is time that fans and consumers realize that for artists it is essential to sell music to keep their heads above water.

Century concluded that:

> Spotify in its present shape and form isn't the way forward …. Ultimately, in some cases, it will completely kill a lot of smaller bands that are already struggling to make ends meet. (Century Media 2011)

More label withdrawals followed. Curiously, metal labels, especially LA-based metal labels, seemed the most keen to remove their catalogues from Spotify. In September 2011, Napalm Records withdrew their repertoire, stating that 'the income generated through streaming is so insignificant that neither we, as a record company, nor our artists are able to further support this method of distribution. We… object to price models that endanger our existence and that of our bands' (see Metal Underground 2011). Also withdrawing in September 2011 were Prosthetic Records and Metal Blade, while Sumerian Records withdrew most of its catalogue in November. Beyond the LA metal scene, a more widely publicised story emerged in November when ST Holdings, a UK digital distributor for independent electronic and dance labels, withdrew from on-demand streaming services the catalogues for 234 of the 238 labels it distributed. In announcing their withdrawal, the company highlighted that, in the third quarter of 2011, their digital revenues dropped for the first time in their history (by 14%) and that iTunes income in particular was badly affected, falling by 24%. During the same period, on-demand streaming services accounted for 82% of 'plays' of their music, but just 2.6% of revenues, with Spotify returning £2500 for 750,000 streams. This low rate led the company to voice concern that the value of music was being undermined, stating that 'music loses its specialness by its exploitation as a low value/free commodity. Quoting one of our labels, "Let's keep the music special, **fuck Spotify**"' (in Resnikoff 2011).

The royalties debate and the changing nature of the musical commodity

It is important to keep these examples in context. The withdrawal of a handful of small independent labels is not going to cause too much of a dent in a catalogue of 30 million tracks.[7] It is also worth noting that some of these companies (such as Century) have subsequently returned their catalogues to Spotify. However, the controversy about the amount of money paid to artists by streaming services has continued, especially among some smaller independents and DIY artists.[8] Broadly speaking, the complaints can be characterised in two ways.

[7] Unless you're a fan of Californian metal or UK dance, in which case it really sucks.

[8] Perhaps most active is David Lowery, member of alternative rock bands Camper Van Beethoven and Cracker who has also worked as a financial analyst and derivatives trader. Lowery runs the blog 'The Trichordist' (sub-

First, musicians argue that on-demand streaming services undermine sales of digital files and physical media and, therefore, they are not a financial model that can sustain their musical careers. Underpinning this, however, is a broader aesthetic and moral argument about what music should be worth, with the micro-pay- ments generated by each stream being seen as devaluing music itself.

I will return to these issues shortly but, first, I want to outline the two ways in which Spotify has responded to such criticisms. The first is to argue that, for many artists, the amount that they receive from Spotify reflects their contract with their record label and not the amount that Spotify pays out (e.g., in Rosenthal 2011).[9] There is some truth to this and it is important to recognise that the question 'how much does Spotify pay artists?' is somewhat misleading given that Spotify does not 'pay artists' any more than a physical record store 'pays artists'. Instead, Spotify pays labels, or distributors handling labels and, as such, what proportion of that money is returned to artists is a matter of individual (though often standardised—see Stahl 2013, 165–170) contracts. It is necessary to be cautious, therefore, of information circulated online about the amount a particular artist received from streaming. However, there is obfuscation in Spotify's response also.[10] Not all of the artists raising concerns have record label contracts, with many instead using digital distributors that charge a flat rate commission which makes it possible to ascertain what Spotify actually paid out. A large number of these independent artists have made significant efforts to provide reliable data on streaming rates (for example, the cellist Zoe Keating and the aforementioned David Lowery, who maintains a 'streaming price bible' on his Trichordist blog). Furthermore, the fact that independent labels, and not just individual artists, have been complaining about the revenue received from on-demand streaming suggests that the low level of income received by artists is not just a matter of exploitative contracts.

Aside from claiming contractual confidentiality, Spotify's main public response to the issue of royalties has been to provide some really big figures—the global amount that it has returned to rights holders. In 2012, Spotify founder, Daniel Ek, stated in an interview that 'the whole debate about artists payments is slightly misconstrued. We have paid out over half a billion dollars to rights holders now, and that's doubled in the last nine months. That's a significant amount of money' (in Dredge 2012).[11] The word 'misconstrued' is quite deliberate because the argument being made is that artists and the media are wrong to treat on-demand streaming services as comparable to sales. Spotify has repeatedly argued that it 'does not sell streams, but access to music. … [It] is not a unit based business and it does not make sense to look at revenues from Spotify from a per stream or other music unit-based point of view. Instead, one must look at the overall revenues that Spotify is generating' (in Houghton 2011).

titled 'artists for an ethical and sustainable internet'), where he provides analytical and provocative posts on a range of music industry issues.

[9] Recently, Spotify has begun to subtly deflect pressure from itself onto the record labels, both by making streaming data directly available to artists (via its 'Spotify Artists' service) and through implicit public comments about the distribution of revenue.

[10] It should also be noted that Spotify's contracts with the major labels are subject to non-disclosure agreements which make it impossible to ascertain how much, and in what ways, specific labels are remunerated. It is commonly accepted that the major labels received lump sum payments for making their catalogues available on the service and that this income is not shared with artists.

[11] In Ek's public response to Taylor Swift, he announced that Spotify has paid $2bn to rights holders (Ek 2014).

Thus, the really really big or the really really small numbers get to the heart of those different ways of understanding streaming services.

To be clear, this is not merely an argument that it is impossible to equate a certain number of streams with one purchase. Rather, it is an argument that the nature of what is being sold is qualitatively different than in the past and, therefore, artists need to reconsider how they are rewarded—not merely how much they are paid but the basis on which they are paid: in the 'old' model of music-buying/owning, consumers paid (and artists therefore received) a one-off payment; in the new model of music-renting, consumers continually pay for access to music. The theory is that continual small payments will compensate (and more) for the withdrawal of the one-off payment. That is why Spotify argues that 'people need to transition from unit-based thinking to consumption-based thinking' (in King 2012).

The economics of streaming music

While Spotify's position reflects a broader cultural shift toward 'renting' music, its argument that musicians should move away from 'unit-based thinking' is seemingly undermined by the fact that musicians are paid on the basis of how many times their songs are streamed by the service. In fact, Spotify royalty statements list each stream and its associated rate (though Spotify refers to calculating per-stream rates as 'reverse engineering'). Understanding Spotify's position that it is wrong to focus on individual streams requires an understanding of the broader economic logic of streaming, which depends upon the totality of streaming rather than the aggregation of individual streams.

Before explaining Spotify's payment system, it is important to recognise that Spotify does not seem to be 'cheating' by paying a lower royalty rate than other digital suppliers. At SXSW in 2012, Spotify's chief content officer, Ken Parks, reported that the company pays out 'in the neighbourhood of 65–70% of revenue to rights holders' (in Peoples 2012a) and it has repeatedly affirmed that it pays 'nearing 70%' or 'approximately 70%'. The 70% figure is important because it is what Apple returns to rights holders for iTunes downloads and it broadly matches historically conventional allocation in the record industry.[12] So while not confirmed precisely, Spotify is at least paying rights holders a broadly conventional rate for the right to use music. It distributes money by dividing its total revenue by the number of streams played on the service. However, the amount of revenue Spotify generates relies mainly upon the number of subscribers it has (plus a little extra from advertising), whereas the number of times music can be streamed is effectively limitless. Thus, when this amount paid is compared to the overall number of songs streamed, the numbers look embarrassingly small—the broad average from the information posted online is about $0.004 per stream (0.4 or 0.5 cents per stream, although streams get different rates of return according to a variety of factors, most notably whether it is played by a subscriber or someone using the free service). Given the tiny figures quoted from musicians on per-stream basis, or the '30,000 streams only earned me $120' stories, it is understandable that musicians might be concerned about the amount of money they are receiving. How can Spotify suggest that streaming is a viable option for artists?

[12] The bulk of that 65%–70% is paid to the rights holder of the sound recording, with 10.5% (of total revenue) paid to owner of publishing rights for mechanical royalties and performance rights.

Spotify's answer to the question is scale. Given the company pays a fixed proportion of revenue back to rights holders then, as Spotify's overall revenue increases so, too, does the amount of money returned to rights holders (70% of a billion dollars is more than 70% of 10 million dollars). This means that scale is key to the viability of Spotify and similar services: the more free users it can convert into paying subscribers the greater the level of return to artists and labels, not because more people are playing their music but because there is a bigger pie to share out and the effective per-stream rate thus increases. This can actually be seen in the royalties data that has been shared online: whereas the average rate per stream was initially in the region of $0.002 to $0.003 per stream (as can be seen in Sam Rosenthal's statement quoted earlier), today it is $0.004 to $0.005. Ek announced in November 2014 that '500 thousand listens on Spotify would pay out between three and four thousand dollars' which is an effective per-stream rate of $0.006 to $0.008 (Ek 2014). The logic of the Spotify model has thus shown signs of working, though it remains to be seen whether services can scale sufficiently to provide a viable means of income for musicians. According to Spotify's artist-in-residence, D.A. Wallach, that particular tipping point is achievable, stating that 'if we can get to the scale of Netflix—which has 20 million subscribers—we estimate we'd be paying out to artists what iTunes is paying out on a year to year basis' (King 2012). [13]

Artists are thus being told that they are wrong to treat streams like sales and that they should adopt a patient, long run view. Yet while large labels with cash reserves can potentially ride out the storm, artists who need money to live on now argue that streaming is undermining their current income by cutting into their digital sales. Both Century Media and ST Holdings made explicit reference to the fact that income from digital sales had fallen dramatically since the emergence of on-demand streaming. Spotify's position, perhaps unsurprisingly, is that streaming does not have a detrimental effect on digital sales. For example, Katie Schlosser, a director of label relations at Spotify, has stated that they have been able to provide labels with 'data that's proving and demonstrating the fact that streaming revenue is additional to actual unit download consumption or physical music sales' (Resnikoff 2013). Perhaps more surprisingly, the argument has been supported by the major record labels. For example, at an IFPI event in London in 2012, then president of Universal Music Group's global digital business, Rob Wells, described the argument that on-demand streaming 'cannibalises' sales as 'absolutely bogus' (Peoples 2012b).

The year 2012 was a good year to claim that streaming does not undermine sales, as overall revenue for the recording industry increased for the first time this century, if only by 0.2% (IFPI 2013). There were also individual instances where streaming and selling seem to have complemented each other well. For example, one of the biggest albums of 2012—Mumford and Sons' *Babel*—became the most streamed album on Spotify in its first week of release ('around eight million listens') while simultaneously recording the highest first week sales for a US

[13] In 'Spotify explained' (2013), the company used 40 million subscribers as their basis for explaining the potential of the service for musicians. Two important provisos need to be made about this argument, however. The first is that it assumes subscription prices remain constant, but there are many commentators who think that, for streaming to really break though into the mainstream, lower price points will have to be introduced (for example, Mulligan 2014d). Second, it assumes that new subscribers are converted from existing free tier users. Spotify has thus far remained consistent in converting 25% of users into subscribers. If, however, the number of free users increases at a faster rate than the number of subscribers, then revenue compared to number of streams could conceivably go down rather than up.

album that year (600,000 units) (Peoples 2012b). Furthermore, there are specific territories where streaming has played a significant role in the local recording industry returning to growth after more than a decade of decline. For example, in Norway (where streaming accounted for 65.3% of recorded music revenues in 2013) the industry grew by 7.2% in 2012 and 10.6% in 2013 (IFPI Norge 2014) while in Sweden (where streaming contributed 71.2% of recorded music revenues in 2013) the recording industry has seen 3 years of consecutive growth (IFPI Sverige 2014).

Given their distinctive characteristics (small markets with high broadband penetration and the home of two of the major streaming services), it is unclear whether these Scandinavian examples are reliable indicators of future trends in much bigger music markets. Most notably, neither country had well-established paid download services prior to streaming services emerging (Mulligan 2014b). This makes them very different from the United States and the United Kingdom, where download services were contributing between 40% and 50% of recorded music income by the early 2010s. And, revenues from download sales declined in the United States for the first time in 2013 (Pham 2014) and the United Kingdom in 2014 (Titcomb 2015). Critics of streaming services were quick to attribute the blame to Spotify (for example, Titcomb's headline was 'UK music downloads in decline for first time as Spotify eats into iTunes sales'), though Ek (2014) contends that to do so is to confuse correlation with causation. Instead, he points to Canada, where download sales were falling before streaming services emerged. It should also be noted that growth in download sales had been flattening out considerably since 2008 (which Mulligan (2014c) attributes to increased spending on apps).

In the long run, it seems reasonable to assume that the changing nature of the musical commodity means that streaming will replace downloading but it is difficult to make predictions about the speed and the nature of the transition as current data is ambiguous. Nonetheless, 'while industry executives initially refused to attribute the early signs of weakness in digital sales to the rise in streaming, in the second half of 2013 many conceded that ad-supported and paid subscription services did seem to be cannibalizing digital sales' (Christman 2014).

Spotify and the mainstream industry

Despite concerns that streaming music could be undermining digital sales, the major record labels (Universal Music Group, Sony Music Entertainment and Warner Music Group) have been very supportive of both Spotify and on-demand streaming more broadly. Many of the bigger independent labels, such as Beggars Group and XL, have also been supportive of the service. The majors' support of Spotify is in stark contrast to the early years of the twenty-first century, during which they were hostile to the emergence of digital music and ultimately had to be bullied by Apple into committing themselves to the digital marketplace (Knopper 2009, 170–177). What has changed in the meantime to effect such differing responses? [14]

Understanding the change in attitude requires understanding the changing contours of piracy in the last 15 years or so. In the early twenty-first century, the record industry's

[14] It should not be assumed that the major labels' reaction to Spotify was uniformly positive. In particular, there was resistance to the idea that access to music should be free and, before launching in the United States, Spotify implemented some limits on the amount of music that could be listened to by non-subscribers. These limits have since been removed, however.

overwhelming concern was that digital files generated massive amounts of piracy which undermined legitimate sales. Initial responses to these developments involved searching for ways in which music could be 'locked' through Digital Rights Management systems and aggressive litigation campaigns, against both companies involved in piracy (such as Napster) and individual downloaders (Marshall 2012). Around 2005, however, the record industry changed its approach to online music consumption. While still publicly criticising piracy and campaigning for improved copyright enforcement, the record industry apparently accepted that it would have to adapt to consumer behaviour. The challenge for the labels was finding a way to incorporate the behaviour of a generation which had grown up with music being free and accessible. This has been Spotify's key selling-point: it has consistently presented itself as offering a platform which 'monetises' existing consumer behaviour. In its response to the withdrawal of the Century and Mute catalogues, Spotify stated that the service:

> was launched out of a desire to develop a better, more convenient and legal alternative to music piracy. Spotify now monetises an audience the large majority of whom were downloading illegally (and therefore not making any money for the industry) before Spotify was available. (in Bruno 2011)

The connection between Spotify and piracy is less than coincidental: Spotify founder Daniel Ek was formerly CEO of uTorrent, the most popular client for bittorrent sharing and the Spotify software itself depends on the peer-to-peer technologies that facilitate widespread file-sharing (when a user requests a song, encrypted data is transferred from other users' computers) (Greeley 2011). Despite this heritage, however, a major selling point of on-demand streaming services is that they are generating income for rights holders out of consumer behaviour that had previously generated none. [15]

However, surely the main reason for the major labels' support of Spotify and similar services is that the 'consumption-based' logic of streaming services actually fits the logic of the pre-digital record industry quite well. The success of the major labels has always depended upon 'consumption-based' rather than 'unit-based' profits; their strategy was to release lots of records in the knowledge that only a small number of them would be successful but that the rewards from a small number of hits would outweigh the losses of the remaining releases. Thus, from the major label's perspective, so long as some records made a profit, it did not matter which records sold a lot and which did not. It mattered to the careers of specific individuals (the artist, say, or the A&R executive that signed them), but not to the label overall, whose success depended upon overall levels of music consumption.

The correlation between the 'consumption-based logic' of the major labels and streaming services can be seen in the way that royalty payments are calculated, which is inherently beneficial to those with large catalogues. Spotify's payments are based on a 'market share' system. Simplifying slightly, the basic logic of the system is that an artist is paid a percentage of Spotify's overall revenue determined by the percentage of overall streams that the artist

[15] There is not space in this paper to evaluate the claim that streaming monetises 'pirate' behaviour. There are certainly examples of countries, most notably Spain, in which rates of piracy have reduced as adoption of streaming has increased. There is also some consumer survey data which also suggests a connection between increasing streaming services and declining rates of piracy (for example, NPD Group 2012). Ek's aforementioned caution about confusing correlation with causation should be noted, however, given that other factors, such as the dramatic rise in web takedown notices and the growth of YouTube, will have played a part in declining piracy rates.

accounted for in a particular month (Spotify 2013).[16] There is thus no connection between an individual listener's payment and their listening preferences. This is beneficial to artists and labels who receive a high number of streams because it means that they receive a high proportion of income regardless of how many paying users actually streamed their music. For example, if a listener pays £10 but listens to only one obscure Finnish rapper for the entire month, the rapper will still only receive a very small proportion of the listener's money as the majority of it will be distributed to those artists at the top of the Spotify charts. Independent and less popular artists would benefit from an alternative system of payment that directly distributed an individual's subscription to only those artists to which the individual had actually listened.

Conclusion

The specific complaints about royalty rates raised by independent musicians and labels reflect, after a period of uncertainty, some significant structural recalibrating of the recording industry. Rather than internet technologies providing liberation from old industry dynamics, what we may be seeing is a consolidation of long-established power structures (as predicted by Bukart and McCourt (2004) though not via the precise mechanisms they suggested; see also Rogers 2013). Furthermore, the introduction of streaming may be making the ground outside of the mainstream even more uncertain given that independent artists and labels do not have the luxury of extensive back catalogues and are much less insulated against declining sales. If streaming is 'cannibalising' sales, therefore, it will be disproportionately impacting upon smaller players. Artists signed to bigger labels can be insulated against the changing rhythms of payment for recorded music—if, say, it takes 5 years of streaming income for an album to recoup its costs rather than 2 years of sales income, those with larger catalogues/financial reserves will be able to weather the storm. However, even if streaming services can scale sufficiently to offset the global decline in incomes from record sales, Spotify's current payment system seems skewed towards the major labels.

Thus, despite the seemingly radical shift from ownership to access-based models, the business model of on-demand streaming is one with which the major labels are familiar and which suits their existing strategies. It is unsurprising that they have been supportive of the new services. Indeed, this support has gone beyond the realm of public statement as the major labels are actually shareholders in Spotify: financial filings reported in 2009 state that the major labels have a 16.5% in Spotify, with Merlin, an aggregator for the biggest independent labels, owning a further 1% (Arrington 2009; Jerrang 2009).[17] Furthermore, in late 2012, Access Industries, owners of Warner Music Group, invested $130 million dollars in Deezer for an undisclosed stake. This, again, is a repeat of earlier industry structures, in which each

[16] The actual equation offered by Spotify is 'Spotify's monthly revenue x (artist's Spotify streams/total Spotify streams) x approximately 70% to rights-holders'.

[17] According to Arrington, Sony owns 5.8%, Universal 4.8%, Warners, 3.8%, EMI 1.9% and Merlin 1%. Presumably, Universal assumed EMI's share when it completed its takeover in 2012. There is dispute over how much the labels actually paid for their stakes: Jerrang contends that it was just €8,800 (implying they received shares in exchange for licensing their catalogues) while Arrington disputes this, claiming inside sources state that the labels paid the same rate as other external investors and thus Jerrang's figure is missing three zeros. It is impossible to verify these figures, but it is commonly accepted that the major labels and Merlin do have a stake in Spotify.

of the majors owned their own CD distribution companies (Negus 1999, 55–60). Needless to say, the fact that the labels have a stake in Spotify has merely added to the suspicion and distrust towards the company and its royalty system. The lack of transparency in the entire system has been something that Spotify has tried to address (through its Spotify Artists portal), but they remain limited in information they can release and streaming royalty statements remain incomprehensible to even the most seasoned analysts.

From this perspective, the concerns raised by artists and independent labels about royalty payments from Spotify seem very similar to arguments made about the record industry in the past and, rather than being a 'disruptive technology' (Barr 2013), Spotify and other streaming music services may actually (or also) be quite conservative technologies. There are historical precedents here. In *Pop Song Piracy*, Barry Kernfeld argues that the history of copyright infringement in the music industry follows a repetitive pattern: new technologies or social practices emerge which undermine music rights holders' established forms of profit-making, who respond by yelling 'piracy!'. The public recognises genuinely transformative new practices from blatant copying, however, and, in such cases, rights-holders end up having to adapt to the new practices yet, in doing so, find ways of incorporating them into their existing profit-making practices (2011, 2–4).

Though the story is not complete, it is relatively easy to fit the story of digital music into such a narrative, with on-demand streaming services being the chapter in which the powerful corporations begin to incorporate the disruptive practices and reassert dominance. The recorded music landscape in the streaming era is beginning to bear many similarities to that of the CD era: financial success depends upon scale and catalogue, the major labels have a stake in music distribution networks, and the vast majority of artists do not make any money.

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